Does gamification improve second language learning outcomes, or only engagement?

Gamification improves **second language learning outcomes**, extending beyond mere engagement.

Does gamification improve second language learning outcomes? N = 17

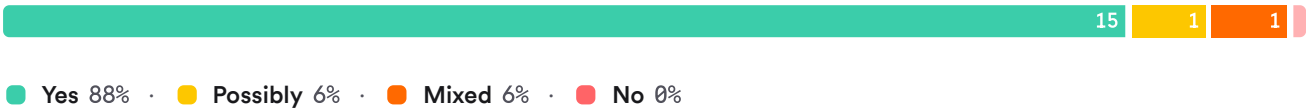


FIGURE 1 Scientific consensus on gamification enhancing second language learning outcomes

Empirical evidence indicates that digital gamification enhances measurable second language outcomes—including vocabulary acquisition, reading proficiency, and grammatical accuracy—rather than acting solely as an engagement booster (Ameen, 2026; Dehghanzadeh et al., 2019; Cheng et al., 2025; Alenazi, 2025; Fatah, 2025). While gamified tools consistently elevate situational motivation and classroom energy (Ameen, 2025; Tsai, 2024), controlled evaluations demonstrate that these affective gains regularly translate into higher test performance compared to traditional instruction (Liu et al., 2024; Zhang & Crawford, 2023; Fatah, 2025; Dorgham & Obiad, 2025).

Measurable Effects on Language Achievement

Quasi-experimental and experimental interventions across multiple target languages confirm that gamification fosters direct cognitive and linguistic improvements (Gutiérrez-Cólon, 2020; Huseinović, 2023; Almelhes, 2024). In vocabulary instruction, game elements like points, badges, and immediate feedback produce substantial gains in lexical acquisition and retention (Ameen, 2026; Alenazi, 2025; Tsai, 2024; Rababah et al., 2025). A trial utilizing a gamified mobile app found that elementary EFL students achieved an overall assessment accuracy of 89.05% compared to 74.01% in the non-gamified control cohort (Zhang & Crawford, 2023). Similarly, semester-long instruction incorporating game mechanics significantly elevated reading comprehension scores among undergraduates relative to lecture-based controls (Cheng et al., 2025).

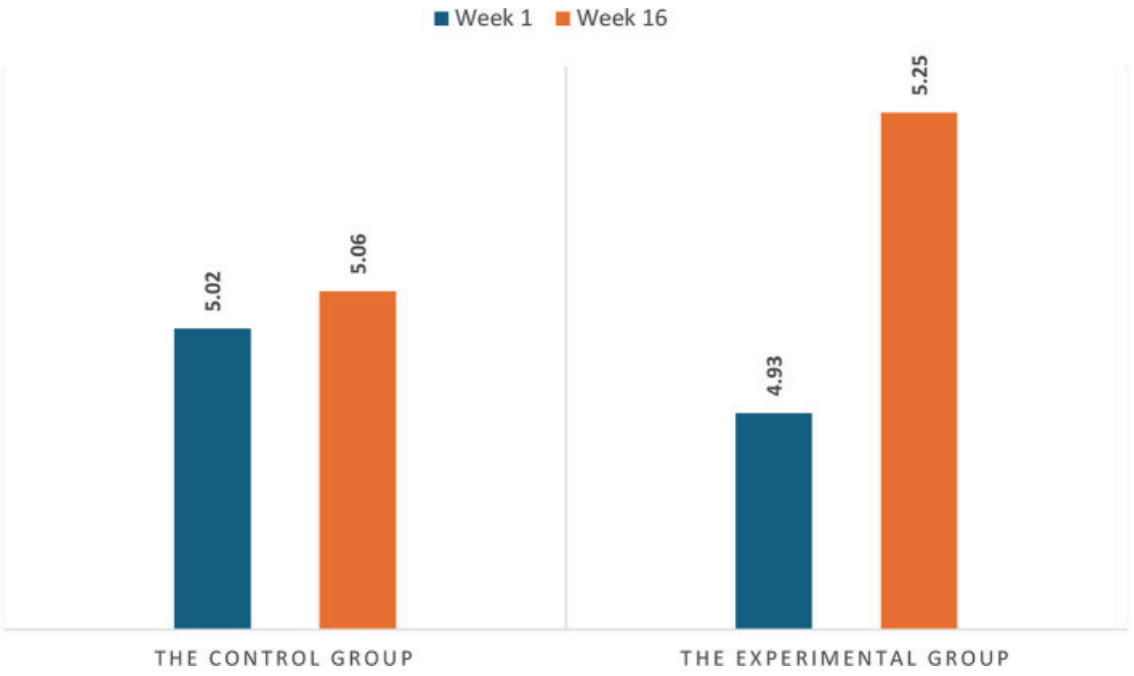


Figure 2. The two groups' reading assessment scores at weeks 1 and 16. (Cheng et al., 2025)
How to read: The bar chart displays reading assessment scores for the experimental and control groups at weeks 1 and 16. Higher bars represent greater reading performance.

Beyond vocabulary and reading, structured gamification yields noticeable enhancements in grammar mastery and spoken production (Cheng et al., 2025; Alenazi, 2025; Dorgham & Obiad, 2025; Almelhes, 2024). Gamified activities provide low-anxiety environments that encourage verbal practice and frequent target language use (La Cruz et al., 2023; Huseinović, 2023; Tsai, 2024). For instance, young learners engaged with gamified management platforms demonstrated marked gains in task completion and oral communication (Ng et al., 2022). Gamified formative quizzes reinforce correct linguistic forms through repeated testing, directly converting active participation into higher post-test scores (Alenazi, 2025; Dorgham & Obiad, 2025).

Comparison of Affective and Cognitive Outcomes

Language Domain	Primary Game Elements	Affective & Behavioral Effects	Assessed Learning Outcomes
Vocabulary learning (Ameen, 2026; Tsai, 2024)	Digital quizzes, flashcards, points (Ameen, 2026; Fatah, 2025)	Higher **study persistence** and enjoyment (Fatah, 2025; Rababah et al., 2025)	Elevated **lexical retention** and delayed recall (Ameen, 2026; Cheng et al., 2025; Rababah et al., 2025)
Grammar acquisition (Cheng et al., 2025; Dorgham & Obiad, 2025)	Immediate feedback, badges, levels (Alenazi, 2025; Dorgham & Obiad, 2025)	Reduced **language anxiety** and fear of errors (La Cruz et al., 2023; Rababah et al., 2025)	Greater **structural accuracy** on exams (Alenazi, 2025; Dorgham & Obiad, 2025)

Language Domain	Primary Game Elements	Affective & Behavioral Effects	Assessed Learning Outcomes
Reading & literacy (Cheng et al., 2025)	Leaderboards, challenges, progress tracking (Cheng et al., 2025; Huseinović, 2023)	Boosted **intrinsic motivation** and teamwork (Cheng et al., 2025)	Superior **reading comprehension** gains (Cheng et al., 2025; Huseinović, 2023)
Oral communication (Cheng et al., 2025; Almelhes, 2024)	Avatars, competitive peer tasks (Huseinović, 2023; Sharifuddin & Abdullah, 2023)	Increased **willingness to speak** (Dorgham & Obiad, 2025; Sharifuddin & Abdullah, 2023)	Improved **pronunciation and fluency** (Huseinović, 2023; Almelhes, 2024)

FIGURE 2 Summary of gamification components and their corresponding affective and linguistic impacts

Caveats and Mediating Conditions

Despite wide reporting of positive gains, several empirical reviews urge caution regarding the depth and longevity of gamified learning effects (Gutiérrez-Cólon, 2020; Luo, 2023):

- **Novelty effects and durability**: Learning spikes often diminish over extended durations as the initial novelty of reward mechanics fades (Ameen, 2026; Ameen, 2025; Ng et al., 2022; Tsai, 2024).
- **Scarcity of delayed testing**: Most published studies rely heavily on immediate post-tests, leaving the long-term cognitive durability of gamified instruction under-evidenced (Ameen, 2025; Ng et al., 2022; Tsai, 2024).
- **Superficial reward mechanics**: Over-reliance on extrinsic point tallies can incentivize superficial gamified guessing rather than deep cognitive processing (Luo, 2023; Alenazi, 2025).
- **Individual learner variation**: Leaderboards and competitive structures energize some students but induce demotivating pressure or unhealthy competition in others (Ameen, 2026; Luo, 2023).

In summary, gamification clearly improves concrete second language outcomes when game mechanics are meaningfully integrated with pedagogical objectives rather than functioning as isolated entertainment (Gutiérrez-Cólon, 2020; Luo, 2023; Alenazi, 2025; Rababah et al., 2025).

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